

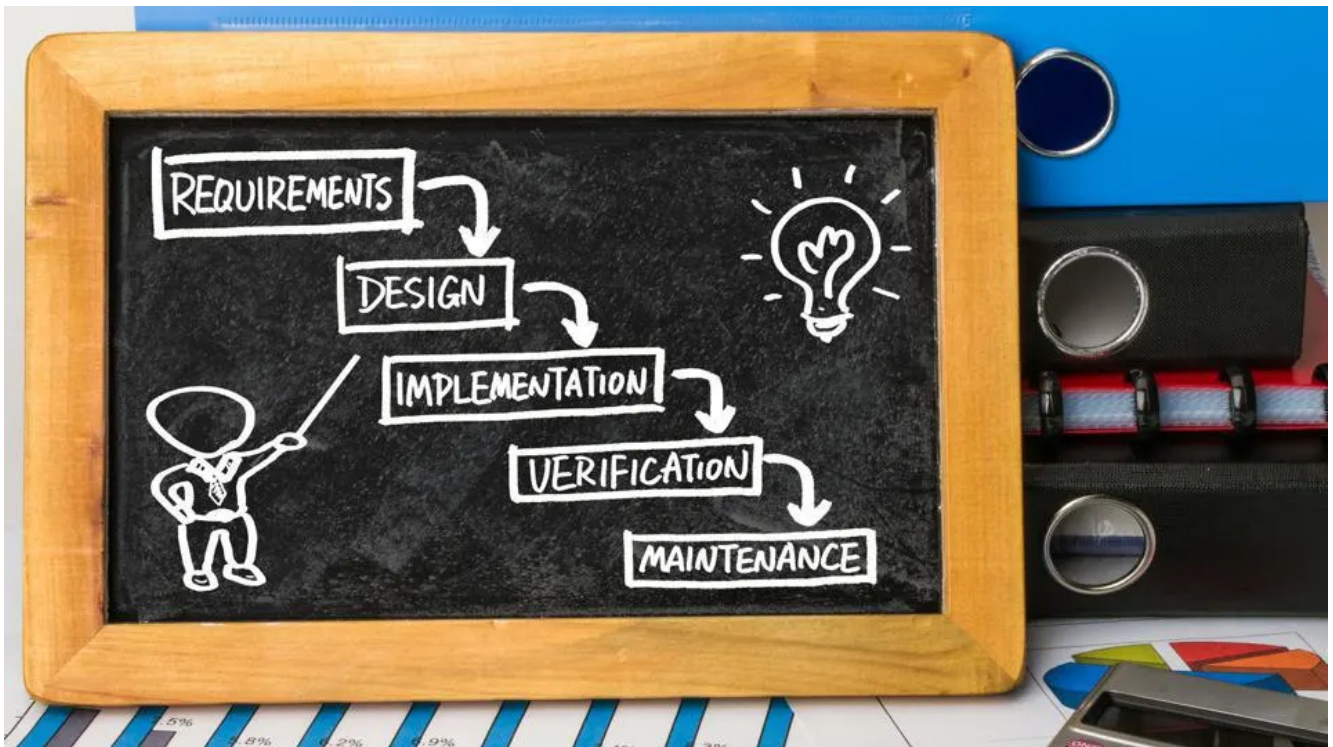
We independently select all products and services. While we earn a commission from partner links, commissions do not affect our editors' opinions or evaluations.

What Is The Waterfall Methodology?

Audited & Verified: Oct 16, 2025, 8:19am

Have a question we may not have answered?

[Ask our editors here](#)



Getty

Table of Contents

- [What Is the Waterfall Methodology?](#)
- [What Are the Waterfall Methodology Phases?](#)
- [Waterfall Methodology Pros and Cons](#)
- [What Industries Use the Waterfall Methodology?](#)
- [Waterfall Project Management Best Practices](#)

The waterfall methodology is one of the most well-known and widely used project management methodologies, especially for predictable projects, such as those in the construction industry. However, if used incorrectly or for the wrong type of project, you will

find yourself questioning your life choices. If you are new to project management and want to know more about the waterfall methodology, you're in the right place.

In this guide, I explain what the waterfall methodology is, how it works and who should use it. I also list some waterfall best practices to help you on your way. As a bonus, I'll even let you know what project management software will help you plan, plot and manage your waterfall project from start to finish.

Featured Partners

Advertisement ⓘ

1 **monday.com**



Learn More →

On monday.com's Website

Free version available

Yes

Starting price

From \$8 monthly per user

Integrations

Zoom, LinkedIn, Adobe, Salesforce and more

2 **ClickUp**



Learn More →

On ClickUp's Website

Free version available

Yes, for unlimited members

Starting price

\$7 per month

Integrations

Slack, Microsoft Outlook, HubSpot, Salesforce, Timely, Google Drive and more

3 **Hubspot**



Learn More

On HubSpot's Website

Free version available

Starting price

\$15 per user

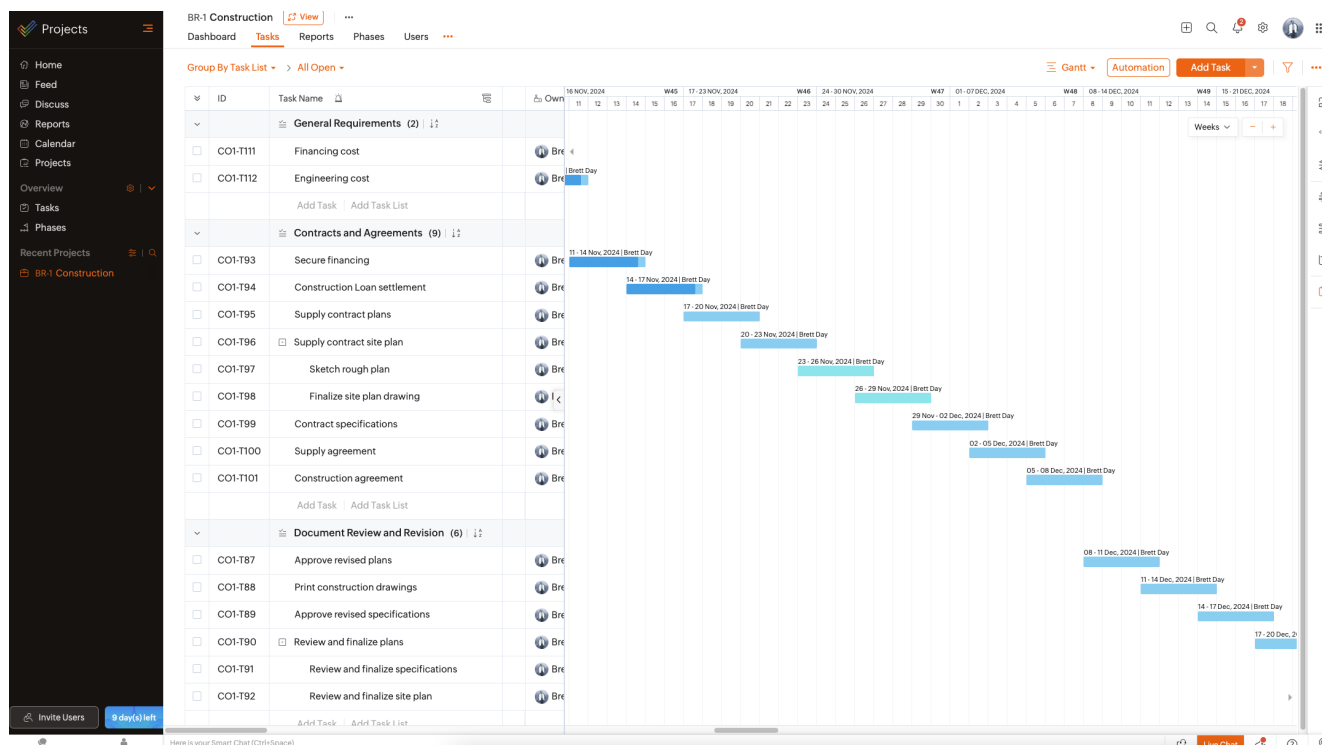
Integrations

Zoo, Wondershare PDFElement, Quip, etc.

What Is the Waterfall Methodology?

The waterfall project methodology is a structured work approach that guides [project managers](#) on a linear path. Waterfall has five distinct phases of work (requirements, design, implementation, verification and maintenance) that contain tasks. Each phase and its respective tasks, when plotted on a [Gantt chart](#), appear to fall down a waterfall. Each phase

must end before the next begins. Waterfall, which is a predictive [project management methodology](#), offers little flexibility but a clear path to the project's goal.



Gantt charts are the preferred tool for those using the waterfall method.

The method, designed by Winston Royce in 1970, was originally created for software development, and to be fair, it worked well for quite some time. However, as the software development industry boomed and markets changed rapidly, waterfall's restrictive methods led to a revolt in the development community, and thus the [Agile Manifesto](#) was created and waterfall's widespread use in software development came to an end. As you'll see later, waterfall is now predominantly used in the construction and manufacturing industries.

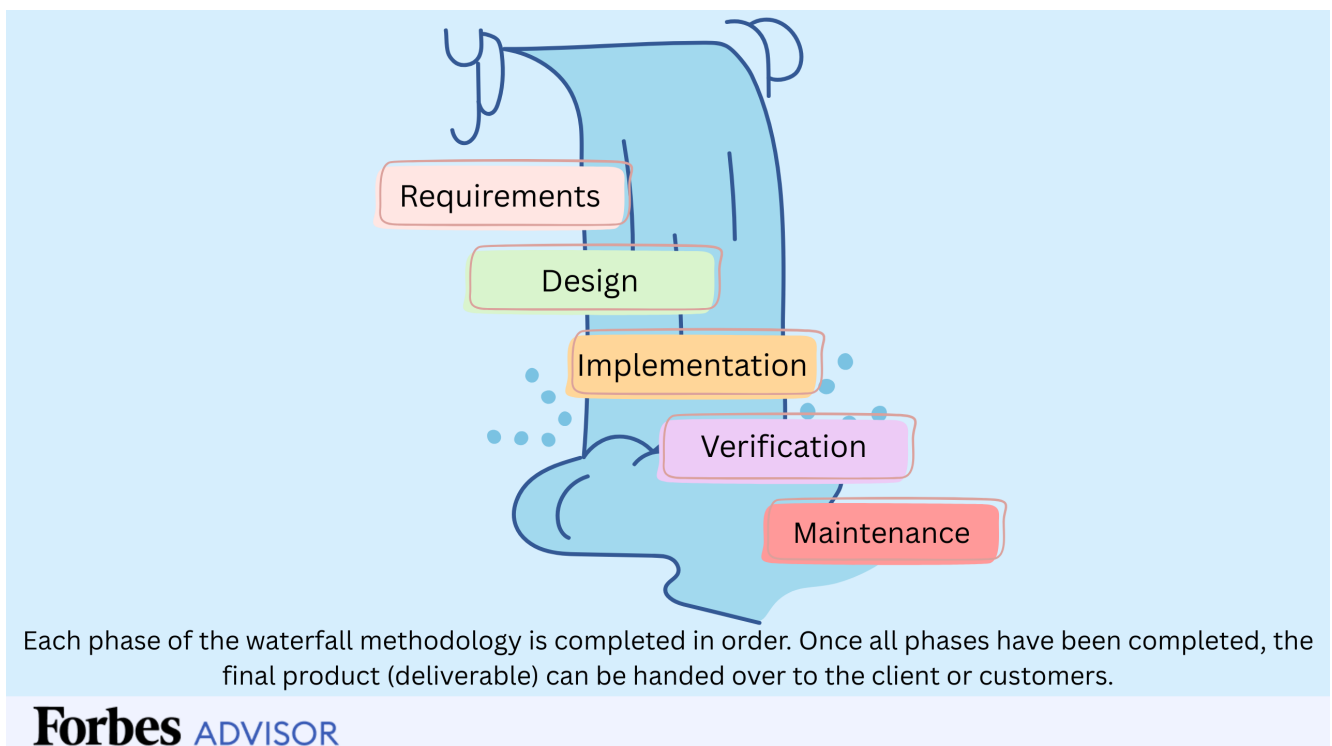
What Are the Waterfall Methodology Phases?

The waterfall project management methodology has five phases, each of which has a set of tasks that must be completed before project managers and their team can move on to the next phase. The five phases of waterfall project management are:

- **Requirements (discovery phase).** During this stage, project managers create a project scope that lists objectives and deliverables (the product), perform a [stakeholder analysis](#), start [resource planning](#), set a timeline, pick team members and detail each stage of the process.
- **Design.** The design phase is when the project team starts to design solutions to meet the end goal. The team sets project schedules and milestones in addition to defining and creating designs for deliverables.
- **Implementation.** As soon as the project team has finished the design phase, they hand off specifications of the deliverables to the team's developers and/or chosen

manufacturers. In addition, the project manager assigns tasks to individuals and [allocates resources](#). The team works diligently through their tasks in sequence until they are complete. Project teams stay focused and communicate very little during this process, an area where waterfall differs from agile frameworks greatly (I discuss this later). This process continues until all tasks and phases are completed.

- **Verification (testing).** Once all phases and tasks have been completed, the project team verifies (tests) the completed deliverable to ensure it meets client or stakeholder requirements.
- **Maintenance.** Once the deliverable has passed the verification stage, the project manager hands over the deliverable to the client or customers so they can use the product. The project team then enters the maintenance phase, which is when after-sales support kicks in. During this time, the team squashes bugs, adds new features or functionalities and offers troubleshoot support.



Each phase of a waterfall project is completed in sequence.

For the most part, the waterfall project methodology works well for projects that have clear deadlines. However, should problems arise, waterfall's structured approach and lack of testing until the deliverable is complete make it hard to correct mistakes and change course. For example, as the next phase cannot start until the current one is complete, any delays can severely disrupt the project's entire timeline.

Casey Halloran, the CEO and co-founder of Costa Rican Vacations, has been leading technical projects for over 25 years. When asked about the waterfall method, Halloran said, "For projects with a fixed scope (for example, an infrastructure upgrade), its linear phases can provide comfort and clarity. The catch is that technology and projects full of curveballs—

waterfall approach often leaves project managers stuck in a bad plan that's not easily able to adapt to a new reality.”

You have to ensure that the project you are working on is compatible with waterfall's fixed approach to working because, just as in real life, it is incredibly hard to climb back up a waterfall if you need to backtrack.

Waterfall Methodology Pros and Cons

As is it with any project management methodology, from [kanban](#) to the [critical path method](#), you will find pros and cons.

Waterfall methodology pros:

- Provides a clear project structure with goals and objectives
- Offers stability for long, complex projects
- Everyone from suppliers to designers is on the same page
- Enhanced cost estimations and timelines can keep [scope creep](#) at bay
- What you see is what you get—the timeline doesn't change

Waterfall methodology cons:

- Extreme rigidity makes it hard to overcome problems
- Communication is limited, meaning it can be hard to share information
- Due to inflexibility, deliverables can have longer delivery times compared to agile
- Defining [S.M.A.R.T. goals](#) and [objectives and key results](#) up front can be hard
- Delays in one phase of the project can have disastrous consequences



Defining S.M.A.R.T. goals at the start of a project with many unknowns can be challenging.

What Industries Use the Waterfall Methodology?

While you still may find some project teams using the waterfall method in the world of software development, you're more likely to find departments of government, construction firms, manufacturing facilities, healthcare and aerospace companies using this traditional method, as the projects in these industries tend to have clear timelines with set goals.

Waterfall Project Management Best Practices

If you're more interested than ever in using the waterfall method but still aren't sure how to make the most of the traditional approach to project management, the best practices I have listed below will help you navigate what can be a complex methodology.

Determine If Waterfall Is the Right Methodology for Your Project

Before diving into the waterfall methodology and trying to make a splash, you must be sure that this approach is the right one for your project. Before starting, you must:

- **Ensure there's a clear path forward**, meaning that you should ensure there will be no major changes to the deliverables and project objectives once the project has started. If there is a chance of this happening, an [agile framework](#) will be better.
- **Meet with stakeholders** and the client to ensure everyone is on board with the plan and the goals of the project. This generally happens during the stakeholder analysis phase. If all is well, be sure to use [RACI charts](#) to list which team members and stakeholders are responsible, accountable, consulted and informed.

- **Be comfortable with little flexibility.** If, for one second, there is any doubt in your mind about the timeline and the potential for changes, choose an agile approach. You're right, I have included this notion twice in this list, as it is that important.

F

Forbesadvisor

Home

Inbox

Chat

Docs

Dashboards

Whiteboards

Forms

Clips

Pulse

Goals

Timesheets

More

Favorites

Spaces

Everything

Team Space

Projects

Production Schedule

RACI Planning

New Website 2025

New Magazine

Create Space

Invite

Help

Team Space / RACI Planning Doc

Share

Link to task or Doc

RACI Planning Doc

Start typing to include a subtitle for your page

Responsible	Accountable	Consulted	Informed
Task owner, accountable for completion.	Decision-maker, responsible for project success.	Provides input/advice, not responsible for completion.	Kept up-to-date, not involved in decision-making.

Project Activity	Product Manager	Designer	Developer	QA Engineer	Content Manager
Define Audience & Goals	R	A	I	C	A
Create Wireframe	C	I	R	I	A
Create Sitemap	A	R	C	R	C
Design Layout	I	C	A	A	I
Develop Website	R	A	I	R	C
Test Functionality	I	R	C	C	R
Launch Website	C	A	I	C	A

Most project management software solutions offer RACI chart templates.

I'm not trying to be an alarmist, but you truly must be comfortable with waterfall's constraints. When used for the right projects, the waterfall approach can lead to greatness. Tom Hamza, a strategic advisor for Trow Avenue Partners, an operations and strategic planning guidance firm, said:

“My biggest success came with a \$200 million enterprise systems rollout for a large auto manufacturer. The program overhauled procurement and vendor integration across more than 100 suppliers, several divisions and extremely tight production schedules. Waterfall's structure—with its clear gates, milestones and documentation—was the only way to keep control of that level of complexity. We went live almost exactly to plan, on time and on budget—which, at that scale, is rare.”

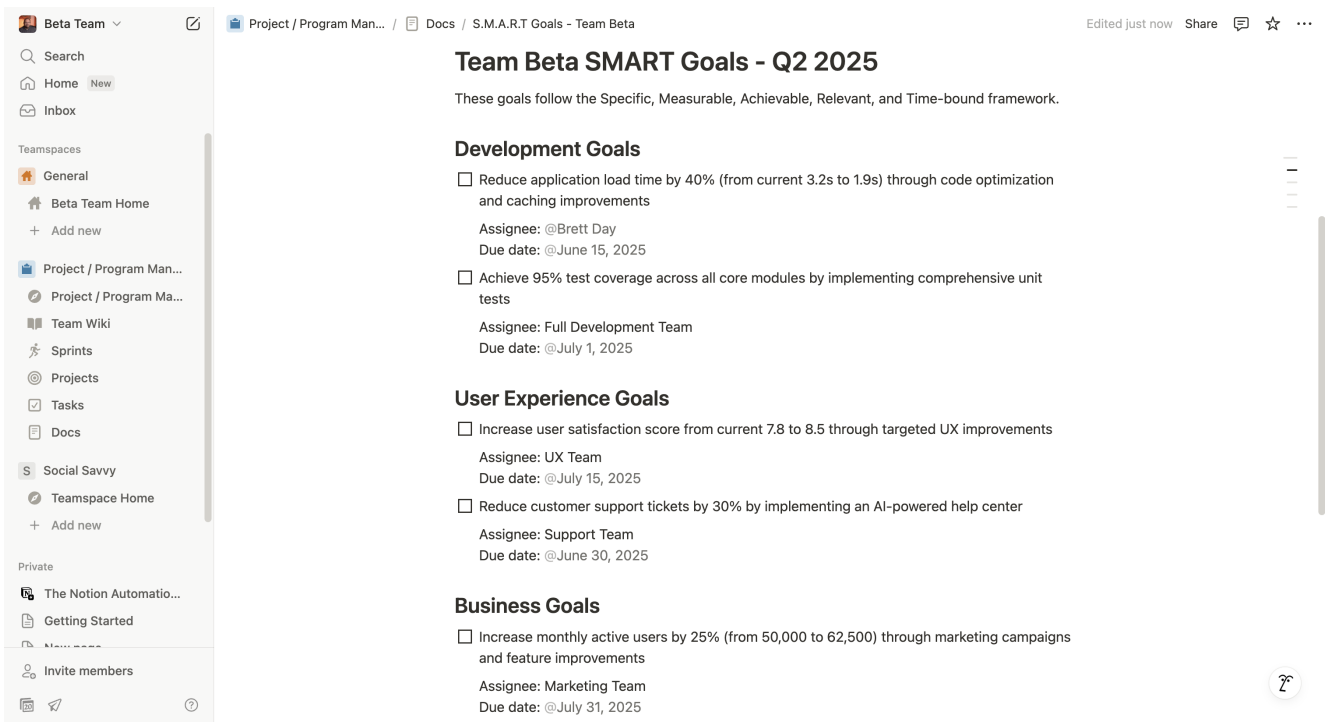
Featured Partners

Advertisement ⓘ

1 monday.com  Learn More → On monday.com's Website Free version available Yes Starting price From \$8 monthly per user Integrations Zoom, LinkedIn, Adobe, Salesforce and more	2 ClickUp  Learn More → On ClickUp's Website Free version available Yes, for unlimited members Starting price \$7 per month Integrations Slack, Microsoft Outlook, HubSpot, Salesforce, Timely, Google Drive and more	3 Hubspot  Learn More → On HubSpot's Website Free version available Starting price \$15 per user Integrations Zoo, Wondershare PDFElement, Quip, etc.
--	--	--

Create Documents for Everything

You will quickly find that waterfall is a document-heavy methodology. From stakeholder and client needs to rigid timelines and resource procurement plans, you must notate everything, even the things that you don't think need to be documented. That way you can refer to them as the project moves forward and protect yourself if/when you get grilled by key stakeholders. I suggest using Notion to make documents. Learn more in our [Notion review](#).



Be as Specific as Possible

Stakeholder Communication

In person

Conference call

Video conference

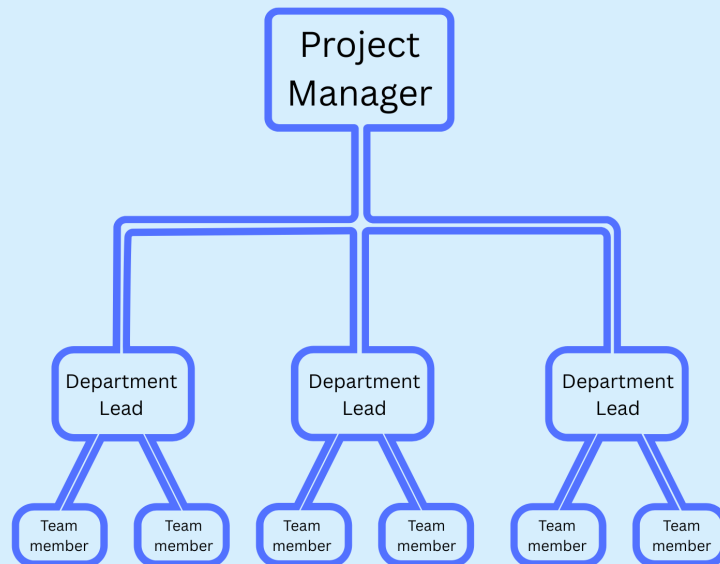
Instant messaging

You can use any communication method to contact stakeholders when conducting a stakeholder analysis. Just be sure to gather as much information as you can.

No matter how you chat with stakeholders, ensure you ask as many questions as possible

Set Clearly Defined Roles

You do not want to head into your project not knowing who will be doing what. Meet everyone on your team to establish a clear chain of command and define project roles. Doing so will get all team members on the same page regarding who to contact if and when issues arise.



Establishing a clear chain of command and defined roles at the start of the project will help everyone understand who to contact when issues arise and what they will be doing.

Forbes ADVISOR

Creating a chain of command can ease communication.

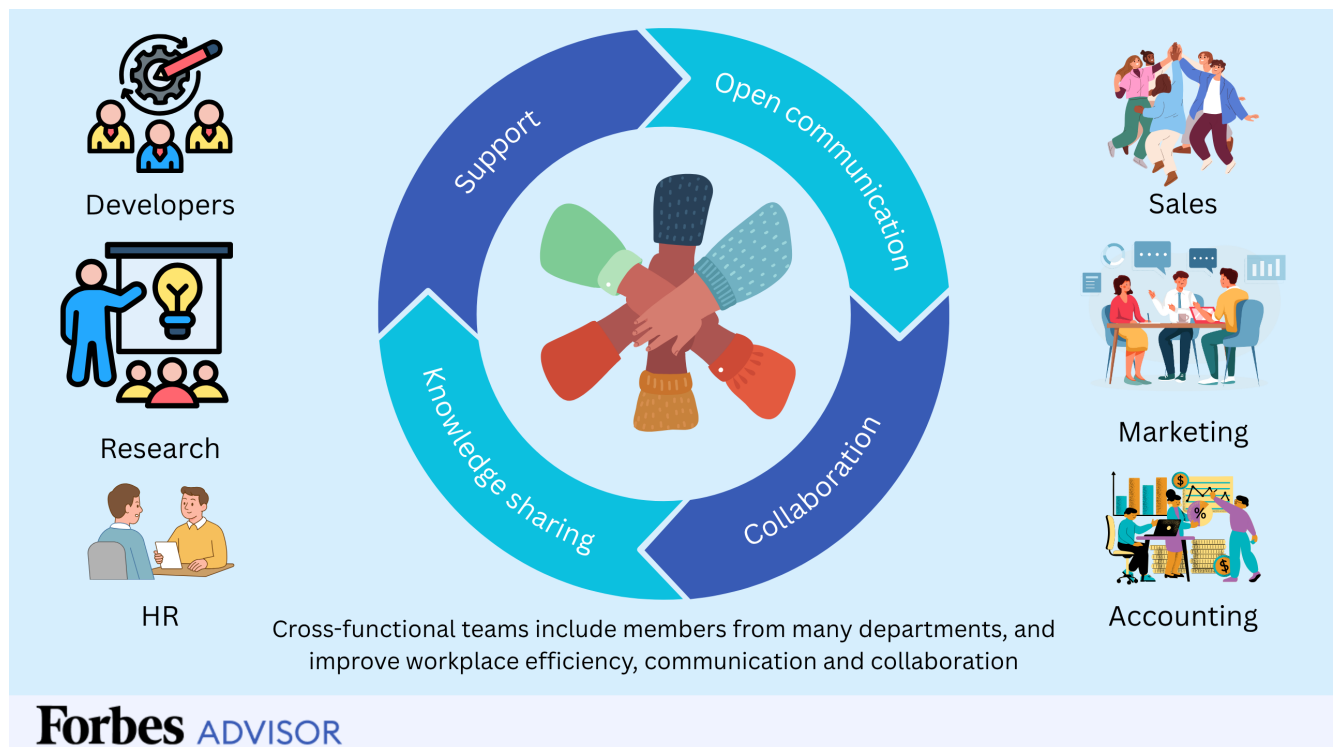
How Does the Waterfall Methodology Differ from Agile?

It's important to note that agile is not a methodology, meaning it does not tell you how to run a project in the way predictive methods, such as waterfall, PRINCE2 and the critical path method do. Predictive methods spell out what needs to be done, when it needs to be done, who can communicate with whom and when the project is expected to be finished. Agile, on the other hand, is simply a list of beliefs and principles that frameworks such as [scrum](#) have been built around. The developers who created the Agile Manifesto favor:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

Jumping from a predictive methodology to an agile framework can be jarring. Gone are the stacks of documents and rigid timelines, as they are replaced with open communication and adaptable schedules that allow you to adjust to changing demands and markets. You'll also

find that workplace silos are replaced with self-managing, cross-functional teams that champion open communication and collaboration.



Cross-functional teams are self-managing and champion open communication.

When Should You Use Agile Instead of Waterfall?

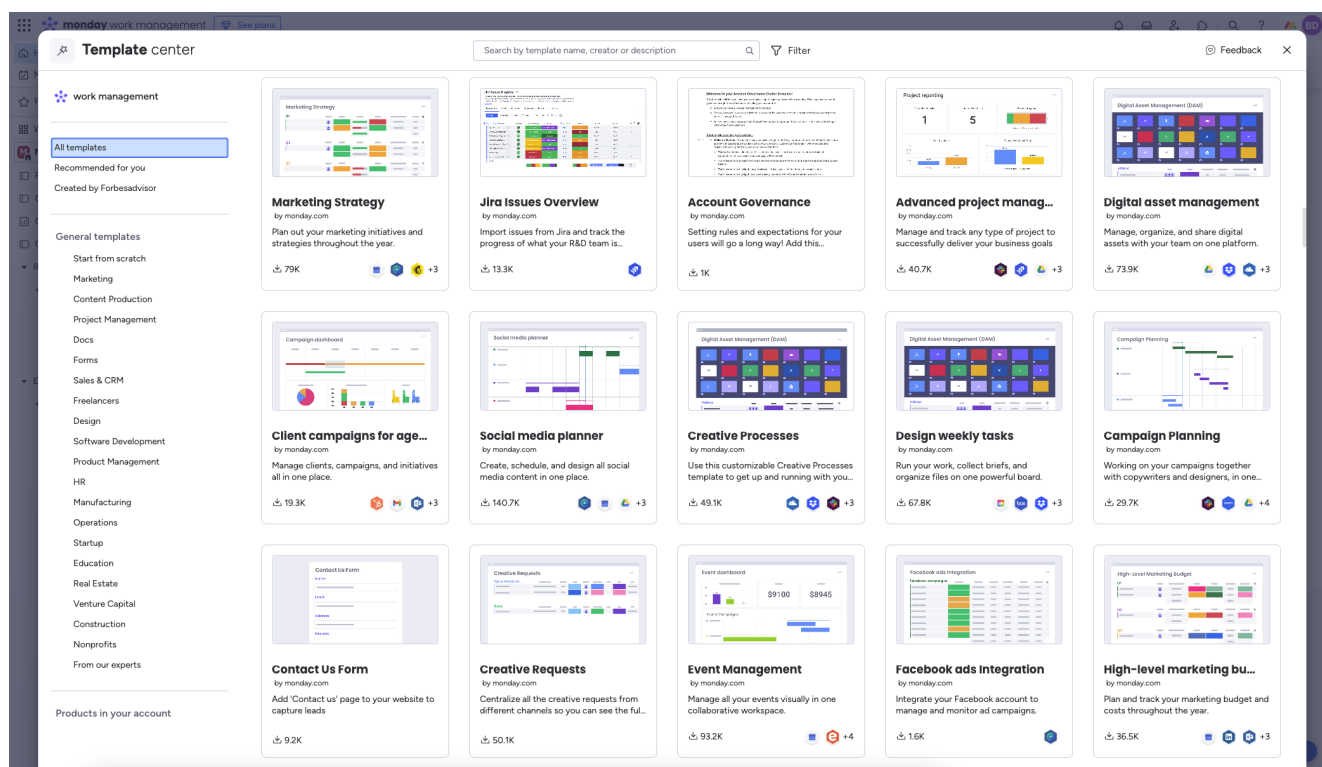
I noted that you should ensure that the waterfall method is right for your upcoming project. If you're in two minds about its suitability, the list below should help you make a decision. Choose an agile framework if you need to:

- **Communicate openly.** Agile champions free-flowing communication across departments. If you need teams to be on the same page every step of the way, an agile approach will suit your project better.
- **React to change quickly.** If the project you're about to undertake is subject to changing client demands and volatile markets, you should choose an agile framework, as timelines, milestones and due dates are not set in stone.
- **Continuous feedback loops.** If your client or key stakeholders need to see working versions of software or the latest iteration of a product regularly to provide feedback, choose an agile framework. Agile ways of working incorporate regular ceremonies or meetings that allow everyone involved to see what's happening, provide feedback and request changes.
- **Collaborate effectively.** If you need to ensure that multiple departments can work together seamlessly, agile is the way to go as self-managing teams can help each other without the need for manager involvement.

The Best Project Management Software for Waterfall

If you have decided that the waterfall methodology is the right one for you but aren't sure what software can help you manage the project, fear not. We spend many hours testing the [best project management software](#) to guide you to the platform that's right for your SMB. Below, I have listed the top three providers that can help you manage a waterfall project.

- **ClickUp.** With intuitive Gantt charts, a robust document maker, mind maps and resource management tools, ClickUp is the best software to use if you're using the waterfall method. On top of the powerful tools, ClickUp offers a feature-dense free plan and affordable paid plans. You can learn more in our full [ClickUp review](#).
- **Monday.com.** If you are new to project management, Monday.com's long list of customizable project templates that are suitable for predictive methods and agile frameworks is for you. Its colorful design and logical menus make navigation a breeze, while the templates can help you get waterfall projects up and running quickly. Learn more in our full [Monday.com review](#).
- **Teamwork.com.** If you have a complex waterfall project coming up, Teamwork.com, with its robust resource management tools and in-depth financial management features, can help you plan and manage your project with ease. You'll find Gantt charts, calendars, lists and even kanban boards to help you keep on top of everything. Learn more in our full [Teamwork.com review](#).



Monday.com's customizable project templates take the guesswork out of creating a project.

Featured Partners

Advertisement ⓘ

1 monday.com  Learn More → On monday.com's Website Free version available Yes Starting price From \$8 monthly per user Integrations Zoom, LinkedIn, Adobe, Salesforce and more	2 ClickUp  Learn More → On ClickUp's Website Free version available Yes, for unlimited members Starting price \$7 per month Integrations Slack, Microsoft Outlook, HubSpot, Salesforce, Timely, Google Drive and more	3 Hubspot  Learn More → On HubSpot's Website Free version available Yes Starting price \$15 per month Integrations Zoom, Slack, Microsoft Outlook, Salesforce, Timely, Google Drive and more
--	--	--

The Bottom Line

The waterfall method is a tried and tested project management methodology that works wonders for those managing predictable projects. Its rigid approach can help set hard goals and deadlines and keep scope creep at bay much more effectively than agile frameworks can. However, the waterfall model has its problems. Should issues arise, project managers will find it hard to adjust the project due to strict deadlines and budgets. Still, when used correctly, the waterfall method is one that can lead a project team to success.

Frequently Asked Questions (FAQs)

- + What is waterfall methodology?
- + What are the benefits of the waterfall methodology?
- + When should you use the waterfall project methodology?

Best Of

[Best Project Management Software](#)

[Best Construction Project Management Software](#)

[Best Project Portfolio Management Software](#)

[Best Gantt Chart Software](#)

[Best Task Management Software](#)

[Best Free Project Management Software](#)

[Best Enterprise Project Management Software](#)

[Best Kanban Software](#)

[Best Scrum Software](#)

[Best Agile Project Management Software](#)

Reviews

[Asana Review](#)

[Trello Review](#)

[monday.com Review](#)

[Smartsheet Review](#)

[Wrike Review](#)

[Todoist Review](#)

[Basecamp Review](#)

[Confluence Review](#)

[Airtable Review](#)

[ClickUp Review](#)

[Motion App Review](#)

Comparisons

[Monday vs. Asana](#)

[Clickup vs. Asana](#)

[Asana vs. Trello](#)

[Asana vs. Jira](#)

[Trello vs. Jira](#)

[Monday vs. Trello](#)

[Clickup vs. Trello](#)

[Asana vs. Wrike](#)

[Monday vs. Wrike](#)

Read More

[What Is Project Management](#)

[Project Management Methodologies](#)

[10 Essential Project Management Skills](#)

[SMART Goals: Ultimate Guide](#)

[What is a Gantt Chart?](#)

[What is a Kanban Board?](#)

[What is a RACI Chart?](#)

[What is Gap Analysis?](#)

[Work Breakdown Structure Guide](#)

[Agile vs. Waterfall Methodology](#)

[What is a Stakeholder Analysis](#)

[What Is An OKR?](#)

Next Up In Project Management

[Project Management Software](#)

[Asana Competitors & Alternatives](#)

[Best Scheduling Apps](#)

[monday.com Review](#)

[Asana Review: Features, Pricing & More](#)

[Trello Review: Features, Pricing & More](#)

More from **Forbes** ADVISOR

Information provided on Forbes Advisor is for educational purposes only. Your financial situation is unique and the products and services we review may not be right for your circumstances. We do not offer financial advice, advisory or brokerage services, nor do we recommend or advise individuals or to buy or sell particular stocks or securities. Performance information may have changed since the time of publication. Past performance is not indicative of future results.

Forbes Advisor adheres to strict [editorial integrity standards](#). To the best of our knowledge, all content is accurate as of the date posted, though offers contained herein may no longer be available. The opinions expressed are the author's alone and have not been provided, approved, or otherwise endorsed by our [partners](#).



Brett Day

Staff Writer

With nearly two decades of retail management and project management experience, Brett Day can simplify complex traditional and Agile project management philosophies and methodologies and can explain the features of project management software in an easy-to-understand manner.
